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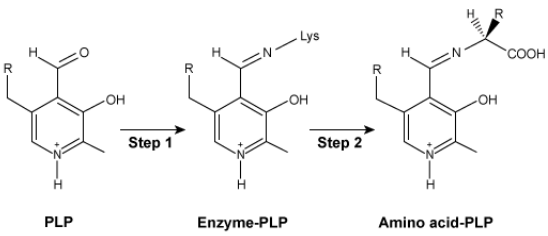


Figure 1 Formation of an amino acid-PLP adduct

The next step in PLP-dependent reactions is the removal of the amino acid's α-hydrogen by a base in the enzyme's active site. A quinonoid intermediate and an additional intermediate are subsequently formed, as shown in Figure 2. The final step includes reformation of the bond between PLP and the enzyme's active site along with the release of the newly formed D-amino acid.

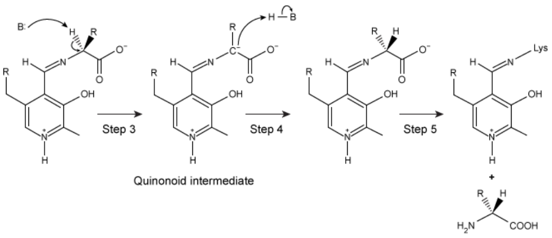


Figure 2 Intermediates of a PLP-dependent amino acid racemase-catalyzed reaction

A few racemases, such as alanine racemase, use a PLP-independent mechanism that proceeds through the enolate intermediate shown in Figure 3.

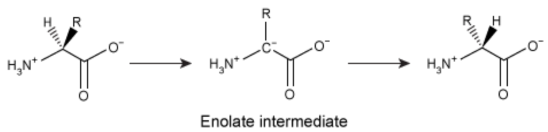
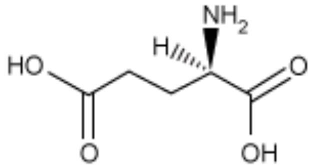


Figure 3 PLP-independent amino acid racemase reaction

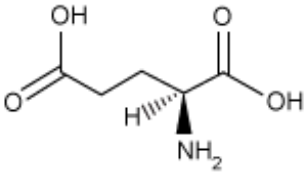
Which of the following structures is the substrate for the reaction catalyzed by glutamate racemase?

A.



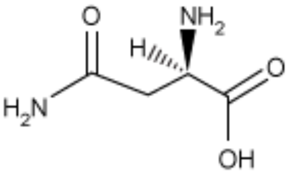
[23%]

B.



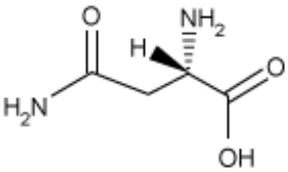
[44%]

C.



[18%]

D.



[14%]

Correct
43% answered correctly

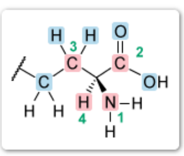
Explanation:

Enantiomers are sets of molecules that differ in their **absolute configuration** around all chiral centers. A chiral center is classified as having an *R* or *S* configuration based on the arrangement of the substituents around the central atom. The α-carbon of all amino acids except glycine is a chiral center and, as stated in the passage, the absolute configuration of most of the *genetically encoded* L-amino acids is an *S* configuration. Of the chiral amino acids, L-cysteine is the only exception, having an *R* configuration due to the high priority of the sulfur atom in its side chain.

The reaction catalyzed by glutamate racemase produces D-glutamate from its *substrate* L-glutamate, which has an *S* absolute configuration. Therefore, the correct answer should show glutamate with an *S* configuration at the α-carbon.

To determine *R/S* configuration:

- Assign priority to each α-carbon substituent.

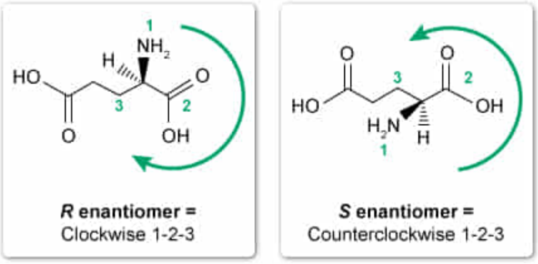


Atoms bonded to chiral carbon
- Nitrogen is highest priority (1)
- Hydrogen is lowest priority (4)
- Carbons are tied

Tiebreaker atoms
- Left carbon (C, H, H) has lower priority (3) than the right carbon (O,O,O) (2)

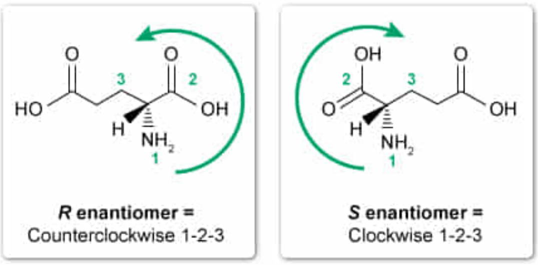
- When the lowest priority group (4) is behind the plane, a clockwise arrangement of the other three substituents corresponds to an *R* configuration, and a counterclockwise arrangement corresponds to an *S* configuration.

Lowest priority group (H)
behind the plane (dash)



- When the lowest priority group is in front of the plane, the opposite is true.

Lowest priority group (H)
in front of the plane (wedge)



The side chain of glutamate, also known as glutamic acid (Glu, E), consists of two straight-chain carbon atoms bonded to a carboxylic acid. The structure in Choice B shows L-glutamate, which has an *S* configuration at the α-carbon.

(Choice A) Since this structure has an *R* configuration, it is D-glutamate, the *product* of the reaction.

(Choices C and D) These structures correspond to asparagine (Asn, N). The side chain of asparagine has one less carbon atom than glutamate and contains an amide rather than a carboxylic acid.

Educational objective:
Of the 19 common L-amino acids, all have an *S* configuration except for cysteine (glycine is achiral). The absolute configuration of *R* or *S* enantiomers is determined by the priority of groups attached to a chiral carbon.

Time spent:0 sec
Subject:
Organic Chemistry

QID:403719
System:
Introduction to Organic Chemistry

A limited number of cellular functions exist for D-amino acids, such as the structural roles of D-alanine and D-glutamate in bacterial cell walls. Because genetically encoded amino acids are synthesized in the L-form, production of D-amino acids depends on enzymes called *racemases*. With the exception of cysteine, conversion of an L-amino acid to a D-amino acid corresponds to the conversion of an *S* stereoisomer to an *R* stereoisomer. The uncatalyzed isomerization reaction requires the formation of a high-energy carbanion intermediate.

Many racemases, such as glutamate racemase, use the coenzyme pyridoxal phosphate (PLP) to form a more stable intermediate. The first step in most PLP-dependent reactions is the condensation of an aldehyde group onto PLP. This forms a Schiff base linking PLP to a lysine side chain on the enzyme's active site. The lysine is then substituted with the amino acid to be converted from the L to the D configuration (Figure 1).

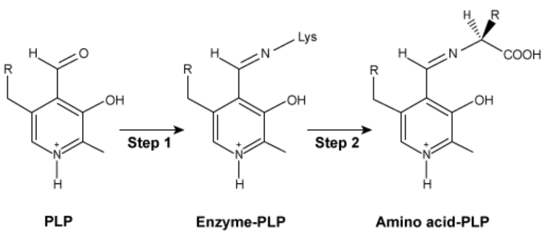


Figure 1 Formation of an amino acid-PLP adduct

The next step in PLP-dependent reactions is the removal of the amino acid's α-hydrogen by a base in the enzyme's active site. A quinonoid intermediate and an additional intermediate are subsequently formed, as shown in Figure 2. The final step includes reformation of the bond between PLP and the enzyme's active site along with the release of the newly formed D-amino acid.

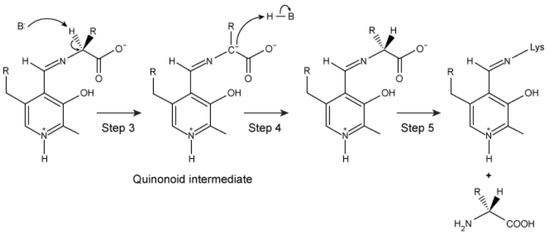


Figure 2 Intermediates of a PLP-dependent amino acid racemase-catalyzed reaction

A few racemases, such as alanine racemase, use a PLP-independent mechanism that proceeds through the enolate intermediate shown in Figure 3.

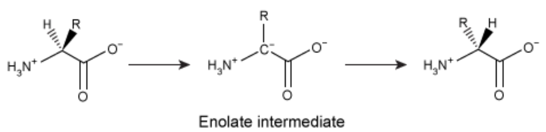


Figure 3 PLP-independent amino acid racemase reaction

The Schiff base shown in Figure 1 contains which of the following functional groups?

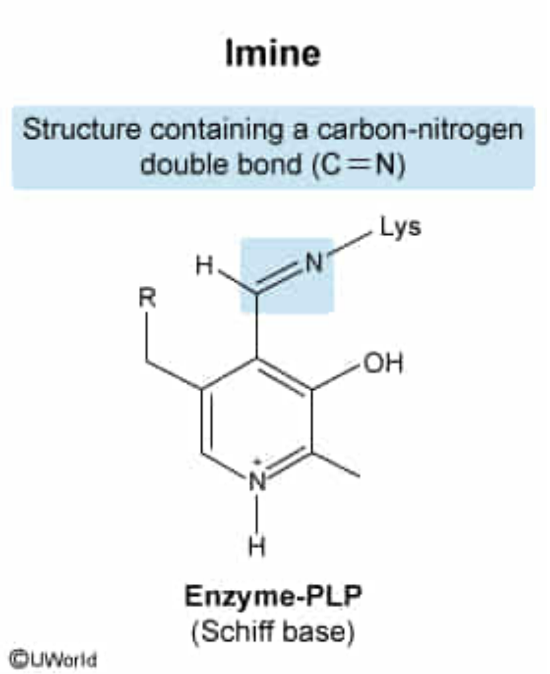
- A. Amide [9%]
- B. Enamine [16%]

○ C. Imide [11%]

✔ ◉ D. Imine [62%]

Correct
63% answered correctly

Explanation:



The structure of an **imine** is comparable to that of a ketone in that they both contain a carbon-heteroatom double bond. Imines are composed of a **carbon-nitrogen double bond** with either a hydrogen atom or an R group attached to the nitrogen atom. Imines that contain an R group on the nitrogen atom are known as Schiff bases.

The Schiff base shown in Figure 1 contains a carbon-nitrogen double bond (C=N) and therefore is an example of an imine.

(Choice A) An amide is a carboxylic acid derivative with a carbonyl carbon atom bonded to an amine group, as opposed to a hydroxyl group.

(Choice B) Enamines are functional groups that contain an amine group bonded to an alkene. Many imines can be converted to enamines through tautomerization. However, tautomerization is not possible in the given Schiff base because no hydrogen atom is available on the carbon adjacent to the C=N to migrate to the nitrogen atom.

(Choice C) An imide is a functional group with a nitrogen atom bound to two acyl groups (ie, two carbonyl carbon atoms). Imides are structurally similar to acid anhydrides but include bonds with the acyl groups instead of oxygen atoms.

Educational objective:
Imines are functional groups composed of a carbon-nitrogen double bond with hydrogen or an organic group attached to a nitrogen atom.

Time spent:0 sec **QID:**403720
Subject: **System:**

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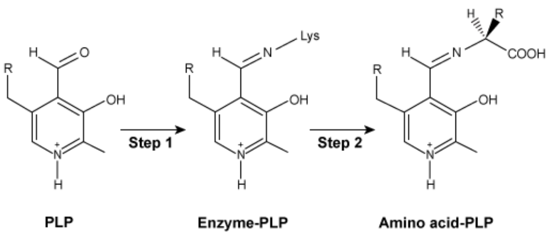

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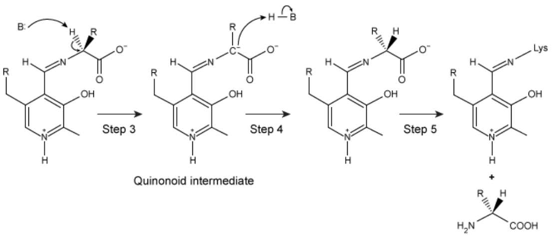

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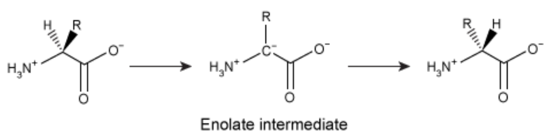


Figure 3 PLP-independent amino acid racemase reaction

During conversion of an amino acid from the L to the D configuration in a PLP-dependent enzyme, a base removes a proton from the α-carbon. Re-protonation of the resulting planar molecule involves the nucleophilic attack of a proton donor:

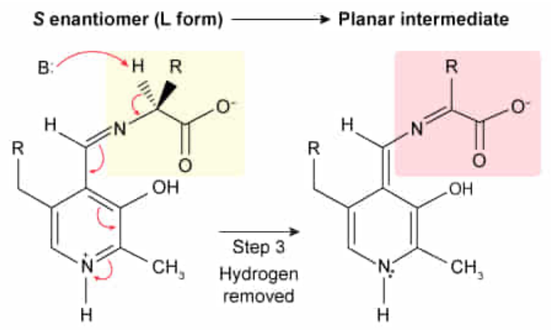
- A. in a nonstereospecific manner. [5%]
- B. on either side of the planar intermediate with equal likelihood. [19%]
- C. with an orientation opposite that of the proton that was removed. [68%]
- D. with the same orientation as the proton that was removed. [6%]

Correct
67% answered correctly

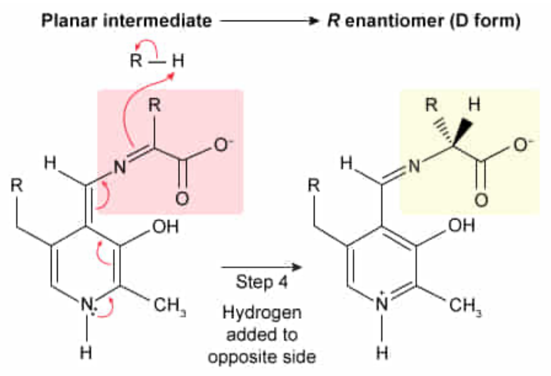
Explanation:

During a **stereospecific** reaction, the configuration of the substrate completely determines the configuration of the product. When one chiral molecule is converted to its enantiomer or to a different chiral molecule, the mechanism often includes formation of a **planar intermediate** such as an alkene or carbocation. The reaction is stereospecific if a new reactant is preferentially added to one side of this intermediate.

For example, the racemase-catalyzed conversion of L-glutamate to D-glutamate is stereospecific. The planar intermediate (imine) is formed in step 3 of the PLP-dependent reactions (Figure 2) after hydrogen is removed from the chiral carbon atom in L-glutamate.



In step 4, the enzyme catalyzes the **addition of hydrogen to** the **opposite side**, which allows for the conversion to D-glutamate.



(Choices A and B) Addition of hydrogen to either side of the planar intermediate would result in the production of a mixture of L- and D-amino acids. The question asks specifically for the mechanism involved in converting L-amino acids to D-amino acids.

(Choice D) The addition of hydrogen to the same side as the hydrogen atom that was removed in step 3 would result in reformation of an L-amino acid rather than formation of a D-

微信Mai_ange

amino acid.

Educational objective:
Conversion of one chiral molecule to its enantiomer often requires the formation of a planar intermediate. The reaction is stereospecific if a side group is preferentially added to one side of this intermediate.

Time spent: 0 sec	**QID**:403721
Subject:	**System:**
Organic Chemistry	Introduction to Organic Chemistry

A limited number of cellular functions exist for D-amino acids, such as the structural roles of D-alanine and D-glutamate in bacterial cell walls. Because genetically encoded amino acids are synthesized in the L-form, production of D-amino acids depends on enzymes called *racemases*. With the exception of cysteine, conversion of an L-amino acid to a D-amino acid corresponds to the conversion of an *S* stereoisomer to an *R* stereoisomer. The uncatalyzed isomerization reaction requires the formation of a high-energy carbanion intermediate.

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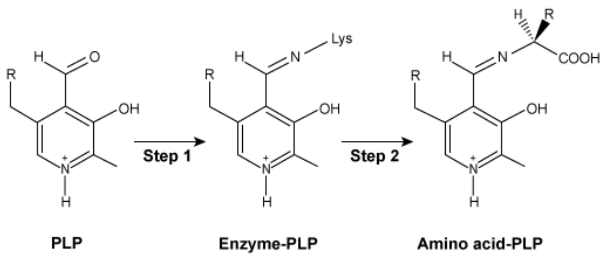

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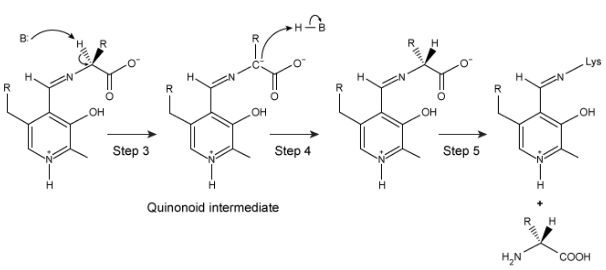
The next step in PLP-dependent reactions is the removal of the amino acid's α-hydrogen by a base in the enzyme's active site. A quinonoid intermediate and an additional intermediate are subsequently formed, as shown in Figure 2. The final step includes reformation of the bond between PLP and the enzyme's active site along with the release of the newly formed D-amino acid.



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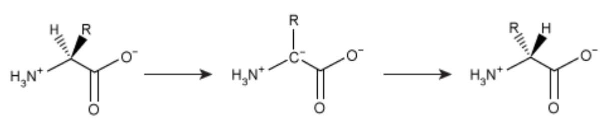
A few racemases, such as alanine racemase, use a PLP-independent mechanism that proceeds through the enolate intermediate shown in Figure 3.



Enolate intermediate

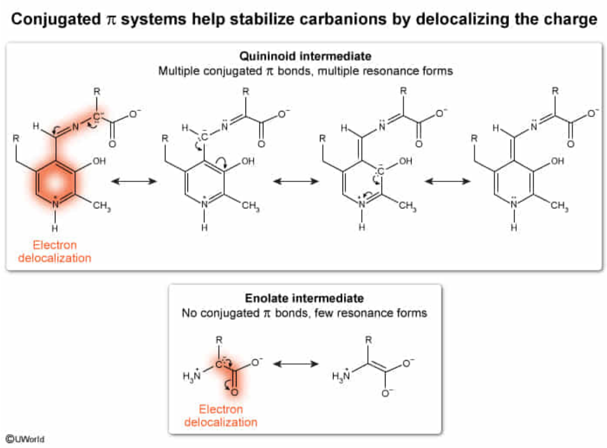
Figure 3 PLP-independent amino acid racemase reaction

Compared to the enolate intermediate shown in Figure 3, the quinonoid intermediate in Figure 2 is:

- A. less stable because of its greater number of resonance forms. [2%]
- B. less stable because of its increased delocalization of electron density. [9%]
- ✔ C. more stable because of its expanded network of conjugated pi bonds. [85%]
- D. more stable because of its smaller number of resonance forms. [2%]

Correct
85% answered correctly

Explanation:



A **conjugated system** can be identified by **alternating single bonds and double or triple bonds**, or in other words, by **alternating pi bonds** (*p* orbitals) separated by sigma bonds. Conjugation allows for the delocalization of electron density, meaning electrons can be distributed through the alternating system's pi bonds. In general, conjugation and electron delocalization serve to stabilize molecules such as carbanions, carbocations, and radicals by creating a more favorable charge distribution. This can be represented as multiple resonance forms of the molecule.

The quinonoid intermediate (Figure 2) delocalizes the negative charge of a carbanion over a large conjugated system of pi bonds, which allows for multiple resonance forms. In contrast, the enolate (Figure 3) delocalizes the negative charge over a relatively small region with fewer resonance forms. Therefore, the quinonoid intermediate is more stable.

(Choices A and D) The quinonoid intermediate has *more* resonance forms than the enolate intermediate, so the quinonoid form is *more* stable.

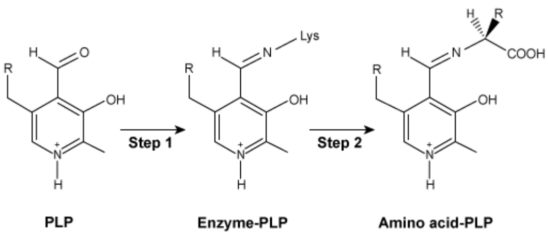
(Choice B) The increased ability of the quinonoid intermediate to delocalize electron density makes it *more* stable than the enolate intermediate.

Educational objective:
A conjugated system is stabilized through electron delocalization via resonance, the distribution of electrons through alternating pi bonds. Conjugated molecules can be identified by alternating single bonds and double or triple bonds.

Time spent: 0 sec	**QID**:403722
Subject:	**System:**
Organic Chemistry	Introduction to Organic Chemistry

A limited number of cellular functions exist for ᴅ-amino acids, such as the structural roles of ᴅ-alanine and ᴅ-glutamate in bacterial cell walls. Because genetically encoded amino acids are synthesized in the ʟ-form, production of ᴅ-amino acids depends on enzymes called *racemases*. With the exception of cysteine, conversion of an ʟ-amino acid to a ᴅ-amino acid corresponds to the conversion of an *S* stereoisomer to an *R* stereoisomer. The uncatalyzed isomerization reaction requires the formation of a high-energy carbanion intermediate.

Many racemases, such as glutamate racemase, use the coenzyme pyridoxal phosphate (PLP) to form a more stable intermediate. The first step in most PLP-dependent reactions is the condensation of an aldehyde group onto PLP. This forms a Schiff base linking PLP to a lysine side chain on the enzyme's active site. The lysine is then substituted with the amino acid to be converted from the ʟ to the ᴅ configuration (Figure 1).



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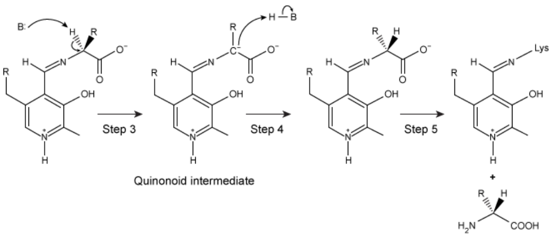

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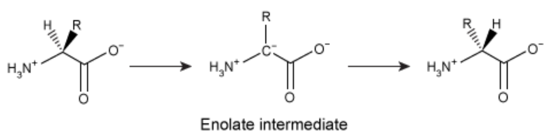
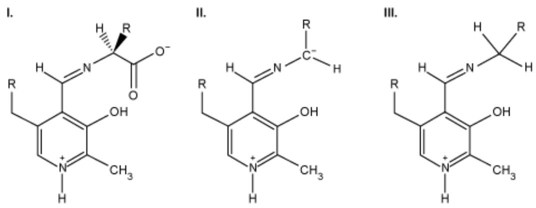


Figure 3 PLP-independent amino acid racemase reaction

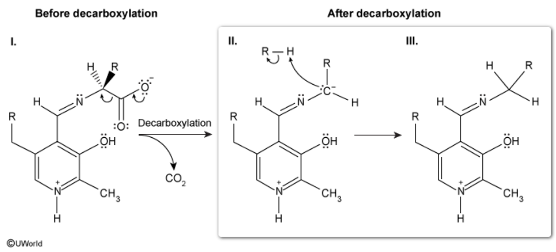
In addition to its role in many racemases, PLP acts as a coenzyme to help decarboxylate certain amino acids by helping stabilize a carbanion. Which of the following structures could be intermediates of this mechanism after decarboxylation has occurred?



- A. I only [12%]
- B. II only [33%]
- C. II and III only [49%]
- D. I, II, and III [4%]

Correct
50% answered correctly

Explanation:



Decarboxylation is the **removal of a carboxyl group** from a compound; the removed carboxyl group is **converted to CO_2**. According to the question, some PLP-dependent enzymes participate in the decarboxylation of α-amino acids by stabilizing a carbanion intermediate **(Number II)**, similar to the mechanism shown in Figure 2. After formation of the carbanion, a proton can be added by a mechanism similar to that of step 4 of Figure 2 to neutralize the negative charge **(Number III)**.

(Number I) This structure has not been decarboxylated.

Educational objective:

Decarboxylation is the removal of a carboxyl group with the release of CO_2.

Time spent:0 sec
Subject:
Organic Chemistry

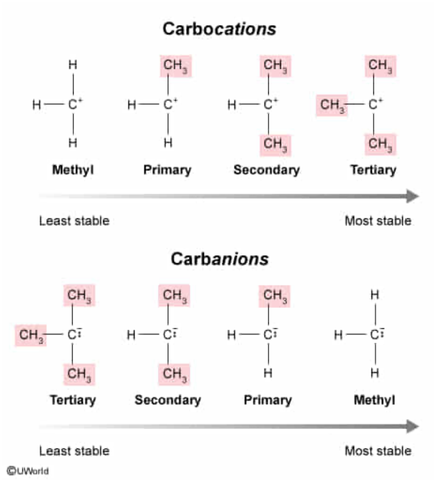
QID:403723
System:
Introduction to Organic Chemistry

Which of the following is true regarding the relative stability of carbocation and carbanion species as the number of alkyl substituents increases relative to hydride substituents?

- ✔ A. As the number of alkyl substituents increases, carbocation stability increases but carbanion stability decreases. [68%]
- B. As the number of alkyl substituents increases, carbocation stability decreases but carbanion stability increases. [12%]
- C. As the number of alkyl substituents increases, carbocation stability and carbanion stability both increase. [16%]
- D. As the number of alkyl substituents increases, carbocation stability and carbanion stability both decrease. [2%]

Correct
68% answered correctly

Explanation:



Carbocations and carbanions are highly reactive species. Carbocations, molecules containing a positively charged carbon atom, are strong electrophiles and a common intermediate in S_N1 and E1 reactions. Carbanions, which contain a negatively charged carbon atom, act as strong nucleophiles due to the lone pair of electrons on the charged carbon.

Because nearby **alkyl groups** are **electron donating**, they help **stabilize carbocations** (ie, make them less reactive) but **destabilize carbanions** (ie, make them more reactive). Additional factors that influence the stability of carbocations and carbanions include resonance and conjugation, hybridization, and inductive effects.

Therefore, the order of decreasing stability for these species is:

- Carbocation: tertiary > secondary > primary > methyl
- Carbanion: methyl > primary > secondary > tertiary

(Choices B, C, and D) Increasing the number of methyl substituents (and decreasing the number of hydride substituents) increases carbocation stability and decreases carbanion stability.

Educational objective:
Because surrounding alkyl groups are electron donors, the order of decreasing stability for carbocations is tertiary > secondary > primary > methyl. Conversely, the order of decreasing stability for carbanions is methyl > primary > secondary > tertiary.

Time spent: 0 sec
Subject: Organic Chemistry

QID: 403724
System: Introduction to Organic Chemistry